EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

STRUCTURAL ISOMERISM

General Classification of Isomerism

1. Two isomeric forms of a saturated hydrocarbon:

(A) Have the same structure

(B) Have different compositions of elements

(C) Have the same molecular formula

(D) All the these are correct

CSI069

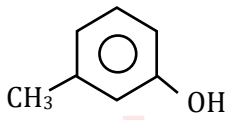
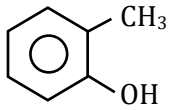
Position Isomerism

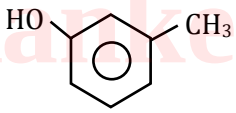
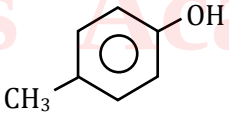
2. Positional isomers of 1-bromo-1-chlorocyclohexane are:

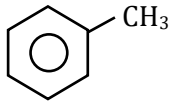
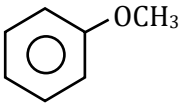
(A) 3 (B) 5 (C) 4 (D) 6

CSI062

3. Which of the following pairs of compounds are not positional isomers?

(A)  and 

(B)  and 

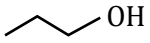
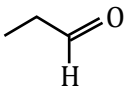
(C)  and 

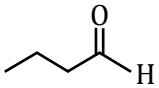
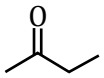
(D) All of these

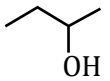
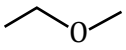
CSI063

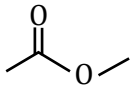
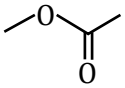
Functional Isomerism

4. Which of the following pairs of compounds are functional isomers?

(A)  and 

(B)  and 

(C)  and 

(D)  and 

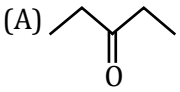
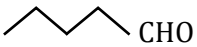
CSI065

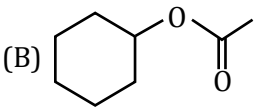
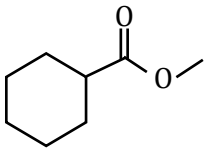
Rankers Academy JEE

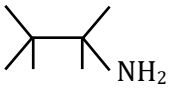
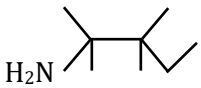
For More Join: @IITJEE_Advanced

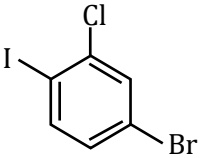
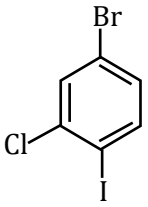
Metamerism

5. Which of the following is a pair of metamers?

(A)  

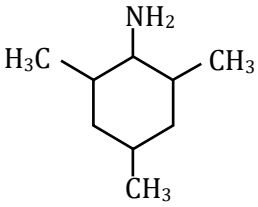
(B)  

(C)  

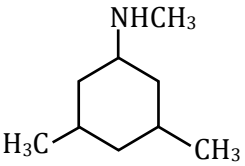
(D)  

CSI066

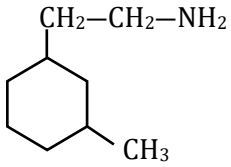
6. Which of the following is **not the CORRECT** relationship?



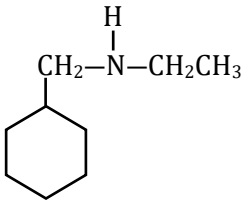
(i)



(ii)



(iii)



(iv)

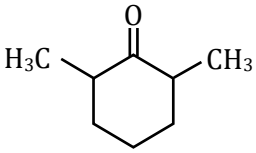
(A) (i) and (ii) are functional isomers
(B) (ii) and (iv) are metamers
(C) (i) and (iv) are positional isomers
(D) (i) and (iii) are chain isomers

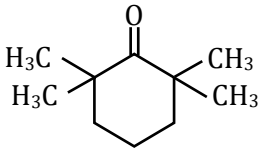
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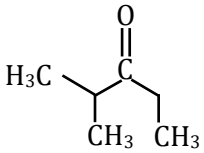
CSI071

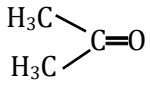
Tautomerism and it's Mechanism

7. Which of the following compounds cannot exhibit keto-enol tautomerism?

(A) 

(B) 

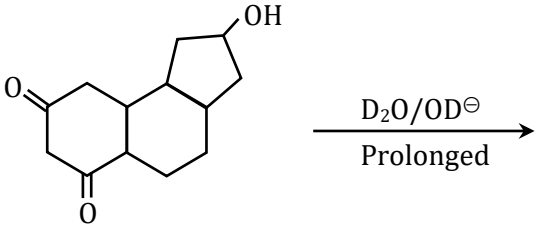
(C) 

(D) 

CSI075

Enol Content and Kinetic Isotopic Effect

8. Number of hydrogen replaced by Deuterium in the given reaction are :



(A) 3 (B) 4 (C) 5 (D) 6

CSI076

For More Join: @IITJEE_Advanced

Identification and Number of Structural Isomers

9. How many different dibromophenols are possible?

(A) 8 (B) 7 (C) 6 (D) 5

CSI067

10. Number of structural isomers of C_3H_9N are :

(A) 6 (B) 5 (C) 4 (D) 3

CSI072

GEOMETRICAL ISOMERISM AND CONFORMATIONS

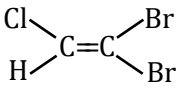
General Definition of Stereoisomerism, Conditions of Geometrical Isomerism in Alkene

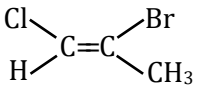
11. Which of the following compound give geometrical isomerism?

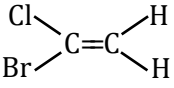
(A) CH_2Cl_2 (B) $CH_2Cl—CH_2Cl$ (C) CHBr = CHCl (D) $CH_2Cl—CH_2Br$

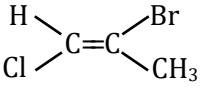
CGI072

12. Which is a pair of geometrical isomers?

(i) 

(ii) 

(iii) 

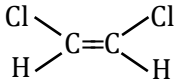
(iv) 

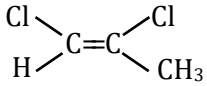
(A) I and II (B) I and III (C) II and IV (D) III and IV

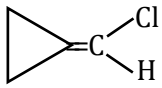
CGI073

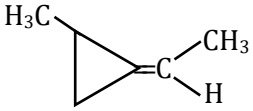
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13. Which Compound don't show Geometrical isomerism:

(A) 

(B) 

(C) 

(D) 

CGI078

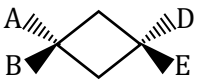
14. When isomers have the same structural formula but differ in relative arrangement of atoms or groups are called

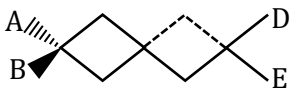
(A) Mesomers (B) Stereoisomers (C) Both A and B (D) None of these

CGI115

Geometrical Isomerism in Allenes, Spiro Compounds, Biphenyls and Triphenyls

15. Which compound does not show G.I. –

(A) 

(B) 

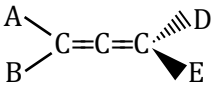
(C) 

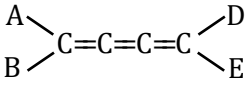
(D) None of these

CGI079

For More Join: @IITJEE_Advanced

16. Which compound show G.I. –

(A) 

(B) 

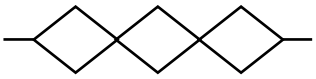
(C) Both (A) & (B)

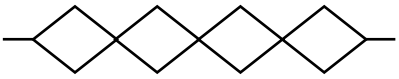
(D) None of these

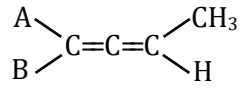
CGI080

17. Which compound show G.I. –

(A) 

(B) 

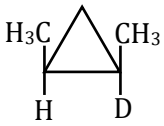
(C) 

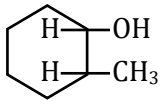
(D) 

CGI081

Geometrical Isomerism in Cycloalkanes and Cycloalkenes

18. Which among the following will show geometrical isomerism :

(A) 

(B) 

(C) $C_6H_5-CH=N-OH$

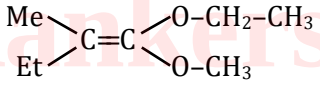
(D) All of these

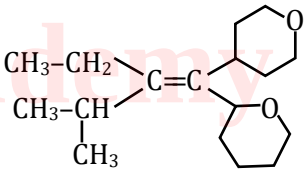
CGI082

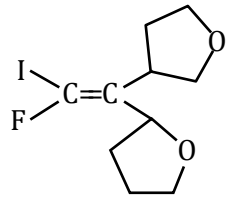
Nomenclature of Geometrical Isomers, Cis-Trans Nomenclature, CIP Rules, E-Z Nomenclature and Syn-Anti Nomenclature

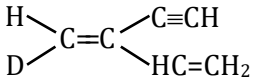
19. Which of the following represent Z-isomer?

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(A) 

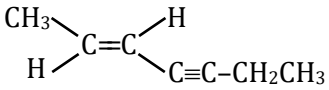
(B) 

(C) 

(D) 

CGI088

20. The IUPAC name of the following compounds is:



(A) (Z) – 5 hepten – 3 – yne

(B) (Z) – 2 hepten – 4 – yne

(C) (E) – 5 hepten – 3 – yne

(D) (E) – 2 hepten – 4 – yne

CGI090

Physical Properties of Geometrical Isomerism

21. Which structure have zero dipole moment

(A) cis-2-butene

(B) 1, 2-dichloro cyclohexane

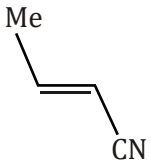
(C) p-dichloro benzene

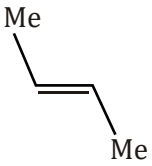
(D) None of these

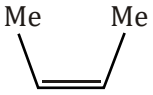
CGI096

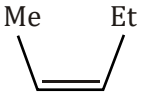
For More Join: @IITJEE_Advanced

22. Which of the following will have lowest dipole moment?

(A) 

(B) 

(C) 

(D) 

CGI097

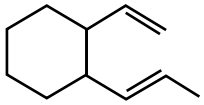
Pseudo Geometrical Isomerism Unit and Calculation of Total Geometrical Isomers

23. The number of geometrical isomers in case of a compound with the structure $CH_3–CH=CH–CH=CH–C_2H_5$ is -

(A) 4 (B) 3 (C) 2 (D) 5

CGI094

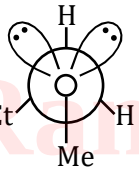
24. The number of Geometrical isomer in case of a compound with the structure

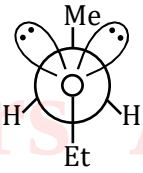


(A) 2 (B) 3 (C) 4 (D) 5

CGI095

Definition of Conformations and Definitions of Gauche, Staggered and Eclipsed

25. 

and


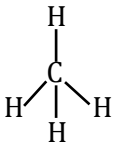
are

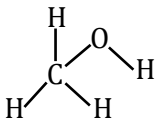
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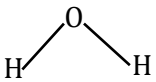
(A) Chain isomers (B) Metamers
(C) Positional isomers (D) Conformers

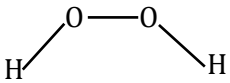
CGI109

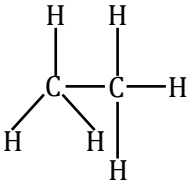
26. Which of the following shows conformational isomerism.

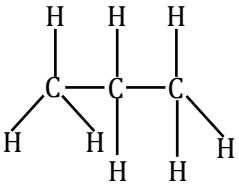
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

(A) b, c, e, f (B) a, b, c, f (C) b, c, d, e (D) b, d, e, f

CGI111

Questions on Stability and Dipole moment

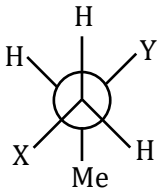
27. Maximum potential energy is associated with which of the following conformers of n-butane:

(A) Anti (B) Gauche (C) Eclipsed (D) Partial eclipsed

CGI105

For More Join: @IITJEE_Advanced

28. Newmann projection representation of $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-Et$ about C_2-C_3 is



Then X & Y are respectively:

(A) Me, Et (B) H, Et (C) Et, H (D) Et, Me

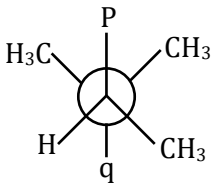
CGI106

29. Bond angle required to acquire most stable form from least stable form in n-butane is :

(A) 360° (B) 180° (C) 120° (D) 240°

CGI114

30. What is the value of p & q in the following conformer of 2,3-dimethyl butane?



(A) p = H, q = H (B) p=CH_3, q = CH_3 (C) p=CH_3, q = H (D) p=H, q=CH_3

CGI116

31. Number of possible conformations of n-butane is:

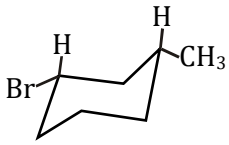
(A) 2 (B) 4 (C) 6 (D) Infinite

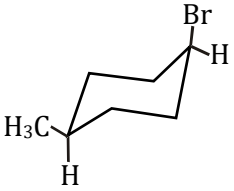
CGI118

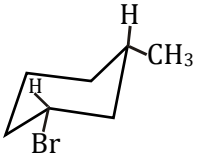
Axial-Equitorial orientation in Cyclo hexane, Ring Flipping; Keq of Flipping; Geometrical Isomerism in Cyclohexane and 1;3 Diaxial Repulsion

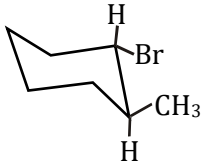
Rankers Academy JEE

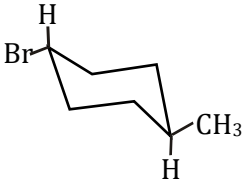
32. How many of the following compounds are cis isomer or a trans isomer.

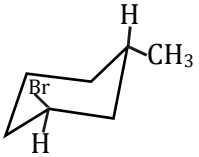
(a) 

(b) 

(c) 

(d) 

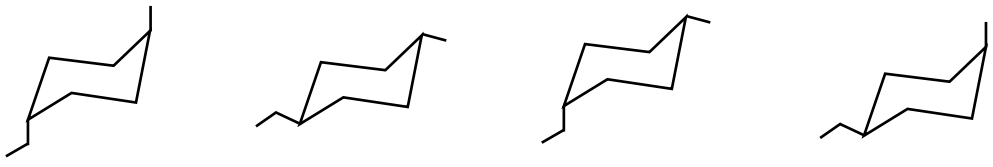
(e) 

(f) 

(A) 4-cis and 2-trans (B) 3-cis and 3-trans (C) 5-cis and 1-trans (D) 2-cis and 4-trans

CGI120

33. Which of the following represent correct stability order of

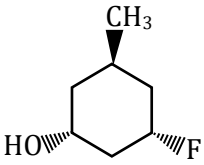


(I) (II) (III) (IV)

(A) II > I > IV > III (B) II > III > IV > I (C) II > IV > III > I (D) II > IV > I > III

CGI123

For More Join: @IITJEE_Advanced

34. 

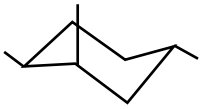
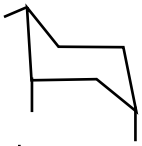
In the most stable conformation of the above compound attached groups OH, F & CH_3 are at respectively :

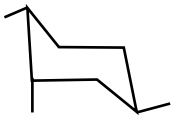
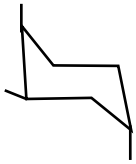
(A) a a a (B) a a e (C) e e e (D) e e a

CGI124

35. Which of the structures (a–d) will be produced if a "ring flip" occurs in the following compound in chair form?

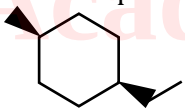


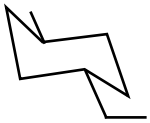
(A)  (B) 

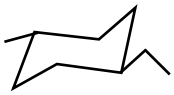
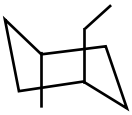
(C)  (D) 

CGI126

36. Which is the most stable chair form of this compound?

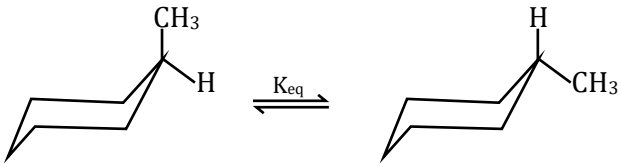


(A)  (B) 

(C)  (D) 

CGI128

37. For given reaction



(A) $K_{eq} > 1$ (B) $K_{eq} < 1$

(C) $K_{eq} = 1$ (D) None of these

CGI130

38. If cis 1, 2-dichlorocyclohexane goes to ring flipping then after ring flipping it will becomes :

(A) cis 1,2-dichlorocyclohexane (B) Trans 1, 2-dichlorocyclohexane

(C) Both cis, trans (D) None of these

CGI131

For More Join: @IITJEE_Advanced

OPTICAL ISOMERISM

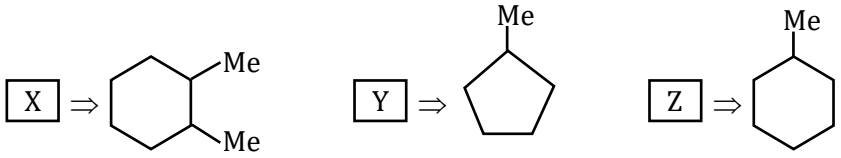
Chiral Compound and Chiral Centre and asymmetric centre and pseudo chiral centre

39. Which of the following compounds is not chiral?

(A) $DCH_2CH_2CH_2Cl$ (B) CH_3CHDCH_2Cl (C) $CH_3CHClCH_2D$ (D) CH_3CH_2CHDCl

COI101

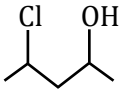
40. Number of chiral carbon atoms in the compound X, Y and Z respectively would be :

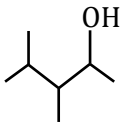


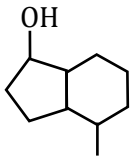
(A) 2, 1, 1 (B) 1, 1, 1 (C) 2, 0, 2 (D) 2, 0, 0

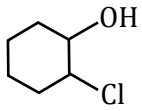
COI102

41. The compound which has maximum number of chiral centres is

(A) 

(B) 

(C) 

(D) 

COI103

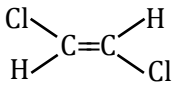
42. Which of the following amine is optically active.

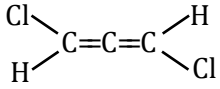
(A) CH_3NH_2 (B) $CH_3NHC_2H_5$ (C) $CH_3CH_2CH_2-N(CH_3)(C_2H_5)$ (D) Sec-Butylamine

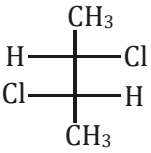
COI121

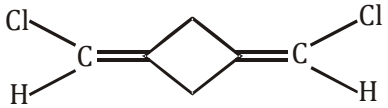
Symmetry Elements: Centre of Symmetry

43. Identify the compound having COS

(A) 

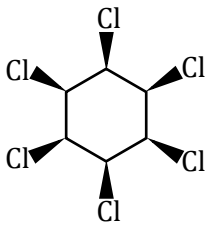
(B) 

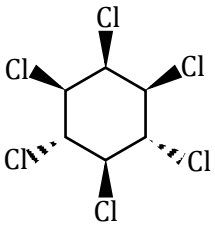
(C) 

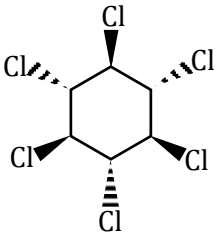
(D) 

COI086

44. Which of the following compound has plane of symmetry and center of symmetry?

(A) 

(B) 

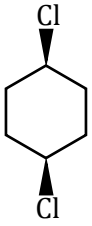
(C) 

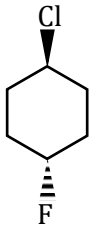
(D) All of these

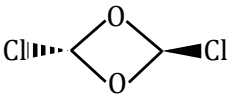
COI087

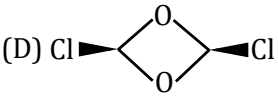
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For More Join: @IITJEE_Advanced

45. Compound having POS and COS both?

(A) 

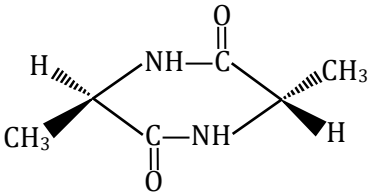
(B) 

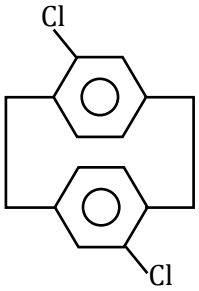
(C) 

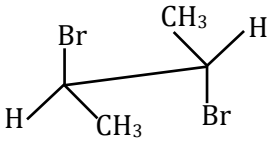
(D) 

COI089

46. Which of the following compounds has centre of symmetry?

(A) 

(B) 

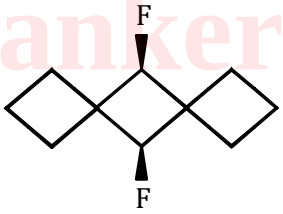
(C) 

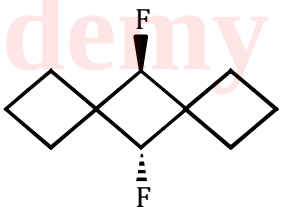
(D) All of these

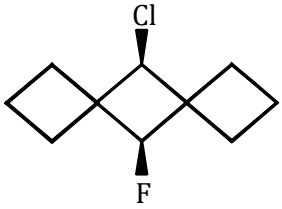
COI092

47. Which of the following containing both POS & COS?

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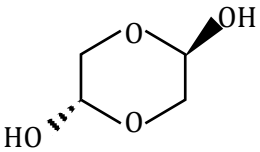
(A) 

(B) 

(C) 

(D) None of these

COI095

48. 

(A)

What is optical rotation of (A) ?

(A) +50° (B) –50° (C) 0° (D) 100°

COI120

Symmetry Elements: Axis of Symmetry; Alternate Axis of Symmetry

49. Which of the following has C_2 & C_3 axis of symmetry?

(A) NH_3 (B) CH_2Cl_2 (C) $CH_3–Cl$ (D) BCl_3

COI090

For More Join: @IITJEE_Advanced

Symmetry Elements: Plane of Symmetry

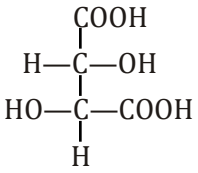
50. Number of POS present in benzene?

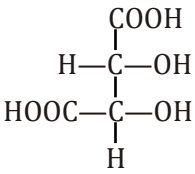
(A) 5 (B) 6 (C) 7 (D) 9

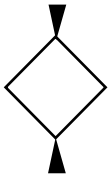
COI091

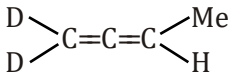
Optical isomerism in allenes, biphenyls, spiro compounds and in rigid body system

51. Identify optically active compound among following :

(A) 

(B) 

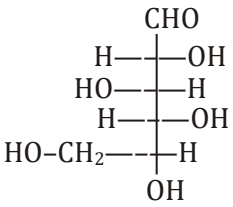
(C) 

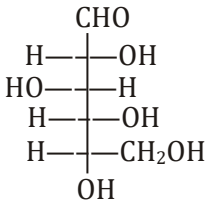
(D) 

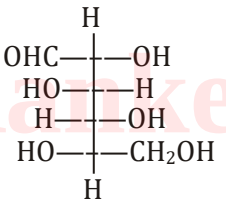
COI100

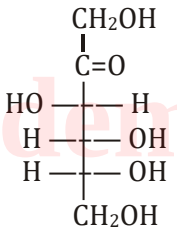
D, L Nomenclature

52. Which of the following is not D sugar :

(A) 

(B) 

(C) 

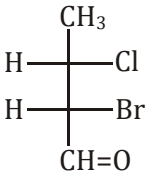
(D) 

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COI109

R, S Nomenclature

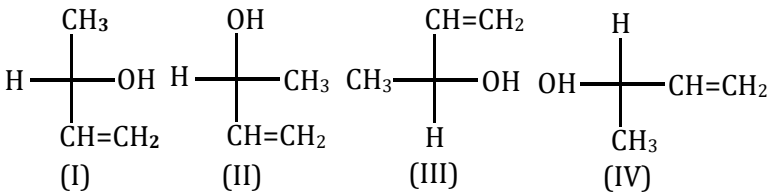
53. The correct nomenclature of following compound is :



(A) 2R , 3R (B) 2S, 3R (C) 2R, 3S (D) 2S, 3S

COI106

54. Which of the following combinations amongst the four Fischer projections represents the same absolute configurations?



(A) (II) and (III) (B) (I) and (IV) (C) (II) and (IV) (D) (III) and (IV)

COI107

For More Join: @IITJEE_Advanced

Optical Purity and Enantiomeric Excess

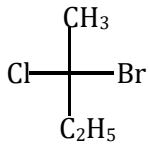
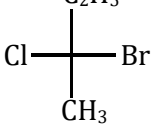
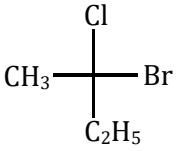
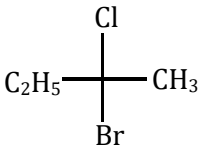
55. What is optical purity of +9° and –1° form ?

(A) 40% (B) 80% (C) 10% (D) 60%

COI119

Assigning relationship in stereo isomers (Enantiomers, Diastereomers and identical isomers)

56. Consider the following structures (A), (B), (C) and (D) –

(A)	(B)	(C)	(D)
			

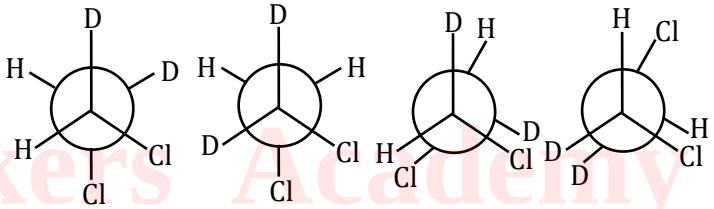
Which of the following statements is not correct

(A) B and C are identical (B) A and B are enantiomers

(C) A and C are enantiomers (D) B and D are enantiomers

COI110

57. The identical compounds are



(I) (II) (III) (IV)

(A) I, II (B) II, III (C) II, IV (D) I, III

COI112

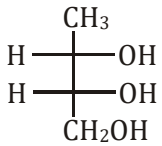
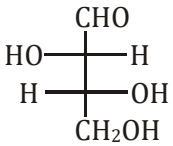
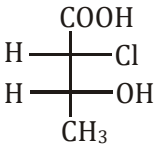
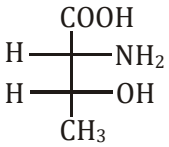
58. The total number of optically active isomer for $CH_2OH(CHOH)_3CHO$ is -

(A) 2 (B) 4 (C) 8 (D) 12

COI125

Erythro and Threo

59. Which of the following is a 'threo' isomer :

(A)  (B)  (C)  (D) 

COI108

Calculation of Optical Isomers and total isomers

60. A compound can be divided into two equal halves & contains even 'n' asymmetric carbon atoms. The number of stereo isomer is -

(A) 2^n (B) $2^{(n-1)}$ (C) $2^{\frac{n}{2}-1}$ (D) $2^{n-1} + 2^{\frac{n}{2}-1}$

COI122

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

STRUCTURAL ISOMERISM

Functional Isomerism

1. Compound having molecular formula $C_nH_{2n-4}O_3$ can have functional group.
(A) 3-Aldehyde group (B) 1-Carboxylic acid & 2-Aldehyde
(C) 1-Carboxylic acid anhydride & 1-Alcohol (D) 1-Carboxylic acid & 1-Alcohol

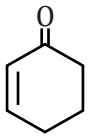
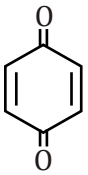
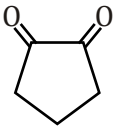
CSI105

Tautomerism and it's Mechanism

2. Correct statements regarding compounds having molecular formula $C_5H_{10}O$ is :
(A) It has four structurally isomeric aldehyde, all can show tautomerism
(B) It has four structurally isomeric aldehyde, out of four three can show tautomerism
(C) It has three structurally isomeric ketones, all can show tautomerism
(D) It has three structurally isomeric ketones, no one can show tautomerism

CSI106

3. Which of the following can show tautomerism ?

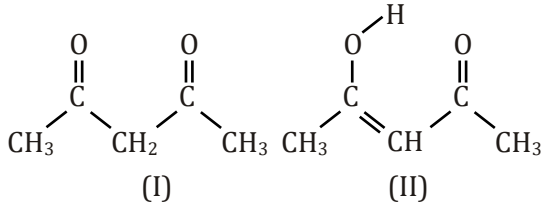
(A)  (B)  (C) $CH_3–CHO$ (D) 

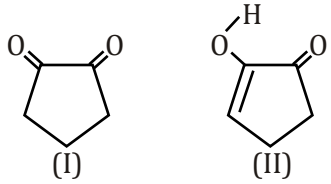
CSI107

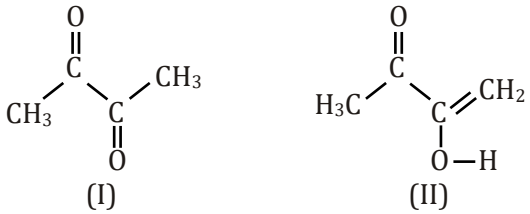
Enol Content and Kinetic Isotopic Effect

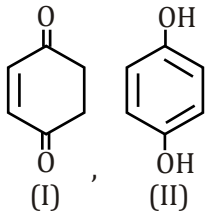
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4. In which of the following pairs 2nd is more stable than its 1st form :

(A) 
(I) (II)

(B) 
(I) (II)

(C) 
(I) (II)

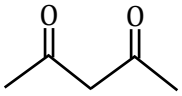
(D) 
(I) (II)

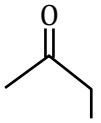
CSI108

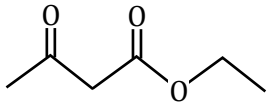
For More Join: @IITJEE_Advanced

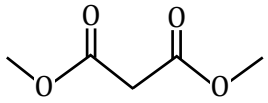
5. Which of the following statements is/are not correct :
(A) Metamerism belongs to the category of structural isomerism
(B) Tautomeric structures are the resonating structures of a molecule
(C) Keto form is always more stable than the enol form
(D) None of these

CSI109

6. Which compound has less enol content w.r. to enol content of compound 

(A) 

(B) 

(C) 

(D) None of these

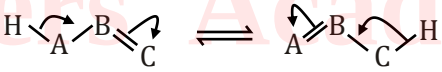
CSI110

COMPREHENSION TYPE QUESTIONS

Paragraph (Q. 7 to Q.8)

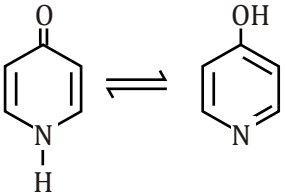
Tautomerism involves interconversion of suitable functional isomers by the 1,3-shift of a hydrogen from an atom which is itself bonded to a functional group containing a double bond, eg : C=O, C=N, N=O and C=C to the terminal atom of the double bond.

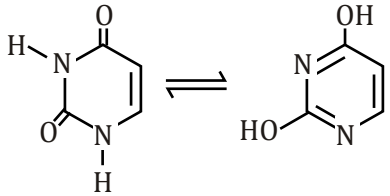
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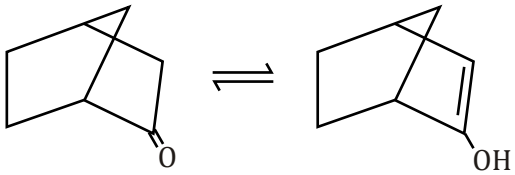
 (prototropy)

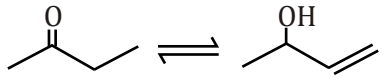
Tautomerism and it's Mechanism

7. Which of the following pair are an example of tautomerism

(A) 

(B) 

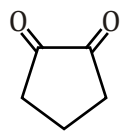
(C) 

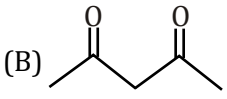
(D) 

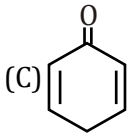
CSI111

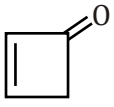
Enol Content and Kinetic Isotopic Effect

8. Which of the following having higher percentage of enol form :

(A) 

(B) 

(C) 

(D) 

CSI112

For More Join: @IITJEE_Advanced

GEOMETRICAL ISOMERISM AND CONFORMATIONS

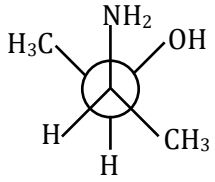
COMPREHENSION TYPE QUESTIONS

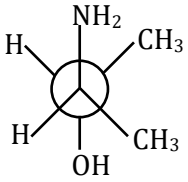
Questions on Stability and Dipole moment

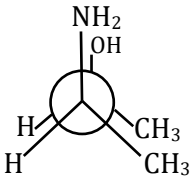
Paragraph for Question 9 & 10

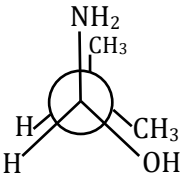
Different special arrangements of atoms that result from free rotation about a single bond are called conformation. Stability of conformer depends on torsional strain, steric strain, dipole-dipole attraction and intramolecular hydrogen bonding.

9. Which one of the following conformers of 3-amino-2-butanol is most stable:

(A) 

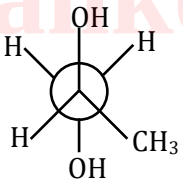
(B) 

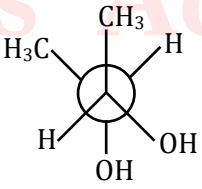
(C) 

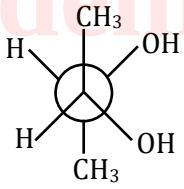
(D) 

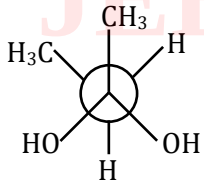
CGI107

10. In which of the following has minimum potential energy in conformers of butane-2-3-diol :

(A) 

(B) 

(C) 

(D) 

CGI108

MULTIPLE CORRECT TYPE QUESTIONS

General Definition of Stereoisomerism, Conditions of Geometrical Isomerism in Alkene

11. For a compound with IUPAC name P_1-methyl pent-P_2-ene which is/are correctly matched. Where P_1 & P_2 are position of substituent & multiple bond respectively.

(A) $P_1 = 2$, $P_2 = 2$ compound doesn't show geometrical isomerism

(B) $P_1 = 2$, $P_2 = 1$ compound doesn't show geometrical isomerism

(C) $P_1 = 2$, $P_2 = 3$ compound shows optical isomerism

(D) $P_1 = 3$, $P_2 = 2$ compound doesn't show geometrical isomerism

CGI140

Geometrical Isomerism in Amines, Oximes and Azo Compounds

12. Which of the following carbonyl compounds can give two oximes on reaction with hydroxyl amine?

(A) HCHO (B) CH_3CHO (C) PhCHO (D) CH_3COPh

CGI141

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Geometrical Isomerism in Cycloalkanes and Cycloalkenes

13. Which of the following is correct statement :

(A) Geometrical isomers are not mirror image isomer.

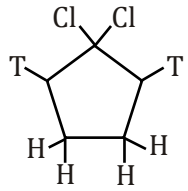
(B) A compound having double bond (restricted bond) always show geometrical isomerism.

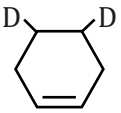
(C) Acyclic compoubd having only single bond does not show geometrical isomerism.

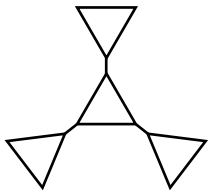
(D) Cyclodecene can show cis & trans form.

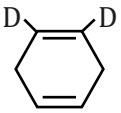
CGI142

14. Which will show geometrical isomerism -

(A) 

(B) 

(C) 

(D) 

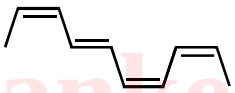
CGI143

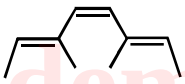
Nomenclature of Geometrical Isomers, Cis-Trans Nomenclature, CIP Rules, E-Z Nomenclature and Syn-Anti Nomenclature

15. Which of the following compounds has cis configuration at each double bond?

(A) 

(B) 

(C) 

(D) 

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CGI144

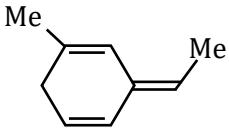
16. Which of the following is true for maleic acid and fumaric acid.

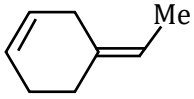
(A) Configurational isomers (B) Stereo isomers

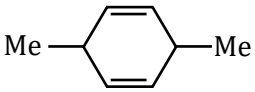
(C) Z and E isomers (D) Constitutional isomers

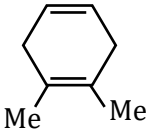
CGI145

17. Which of the following(s) show(s) geometrical isomerism?

(A) 

(B) 

(C) 

(D) 

CGI146

Questions on Stability and Dipole moment

18. For a compound 1, 2-dichloro ethane which of the following statement are correct.

(A) Anti-conformation is most stable one

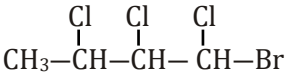
(B) A Gauche conformation has non-zero dipole moment

(C) Eclipsed conformation has plane of symmetry

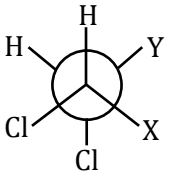
(D) None of these

CGI147

For More Join: @IITJEE_Advanced

19. 

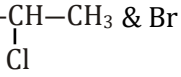
For the above compound (A) the given newmann projection can be represented

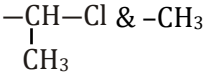


X & Y are respectively:

(A) CH_3 and CHClBr

(B) CH_3 & CH_3

(C) 

(D) 

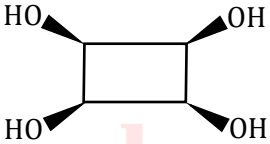
CGI148

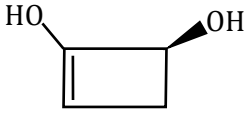
OPTICAL ISOMERISM

MULTIPLE CORRECT TYPE QUESTIONS

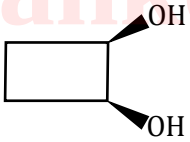
Symmetry Elements: Plane of Symmetry

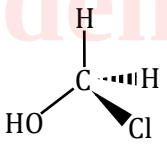
20. Which of the following compound have plane of symmetry?

(A) 

(B) 

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(C) 

(D) 

COI097

Symmetry Elements: Axis of Symmetry; Alternate Axis of Symmetry

21. The number of molecule that posses C_2–axis of symmetry?

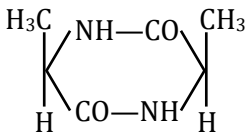
(A) BF_3 (B) Benzene (C) cis-2-butene (D) cyclopropane

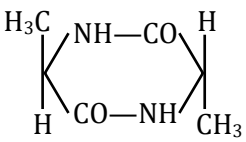
COI096

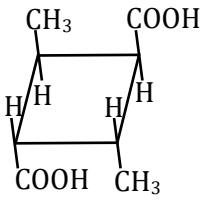
Optical Resolution

22. Which of the following are not resolvable ?

(A) 2,3-Pentadiene

(B) 

(C) 

(D) 

COI126

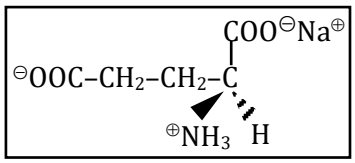
For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Optical Purity and Enantiomeric Excess

Paragraph for Question 23 to 25

S(+) Mono sodium Glutamate (MSG) is a flavour enhancer used in many foods. Fast foods often contain substantial amount of MSG and is widely used in Chinese food. If one mole of above MSG was placed in 845 ml solution and passed through 200 mm tube, the observed rotation was found to be + 9.6°.



23. Find out the specific rotation of (–) MSG :

(A) + 24° (B) +56.8º (C) –48° (D) None of these

COI083

24. Find out the approximate percentage composition of (–) MSG in a mixture containing (+) MSG and (–) MSG whose specific optical rotation is –20° :

(A) 83.3% (B) 16.7% (C) 91.6% (D) 74%

COI084

25. If 33.8 g of (+) MSG was put in 338 ml solution and was mixed with 16.9g of (–) MSG put in 169 ml solution and the final solution was passed through 400 mm tube. Find out observed rotation of the final solution:

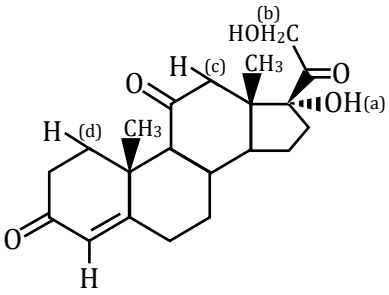
(A) +1.6° (B) +4.8° (C) +3.2° (D) None of these

COI085

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Calculation of Optical Isomers and total isomers

Paragraph for Question No. 26 to 27



Cortisone

26. Total number of chiral centers in Cortisone :-

(A) 4 (B) 5 (C) 6 (D) 7

COI116

27. Total number of possible stereoisomers of the compound Cortisone will be :

(A) 32 (B) 64 (C) 66 (D) 128

COI117

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EXERCISE – (S)

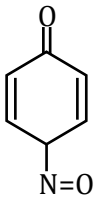
Part-1 : JEE Main (Numerical Value type)

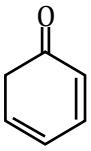
- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

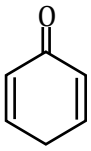
STRUCTURAL ISOMERISM

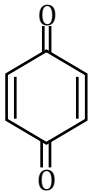
Tautomerism and it's Mechanism

1. Number of compounds which undergoes Tautomerisation to from aromatic product are :

(A) 

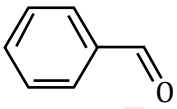
(B) 

(C) 

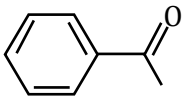
(D) 

CSI113

2. Among the following how many of then will show keto enol tautomerism ?

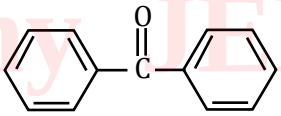
(1) 

(2) CH_3CHO

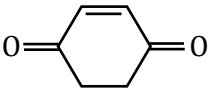
(3) 

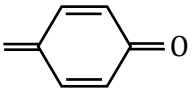
(4) HCHO

(5) CH_3COCH_3

(6) 

(7) CH_3CH_2CHO

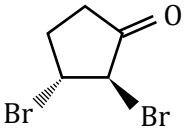
(8) 

(9) 

CSI085

Enol Content and Kinetic Isotopic Effect

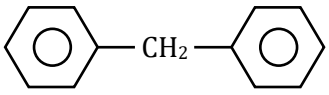
3. How many hydrogen in the compound below are replaced by deuterium when it is shaken in D_2O containing trace of hydroxide ion?



CSI083

Identification and Number of Structural Isomers

4. The molecular formula of diphenyl methane



is $C_{13}H_{12}$

How many structural isomers are possible when one of the hydrogen is replaced by a chlorine atom?

CSI077

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5. How many 5 membered parent chain alkane are possible for C_7H_{16} ?

CSI078

6. Total number of possible structural isomers of $C_5H_{11}Br$.

CSI079

7. How many structural isomer are possible for $C_4H_{10}O$ (only alcohol).

CSI081

8. Number of structural isomer of C_6H_{14} is.

CSI082

9. How many total number of structural isomer are possible for C_4H_7Cl having parent chain of four carbon.

CSI022

10. How many structural isomers will be possible for molecular formula C_5H_{10} (only cyclic) where cyclopropane ring is parent carbon system?

CSI025

11. Total number of esters and acids possible for the compound having molecular C_4H_8O (only structural)?

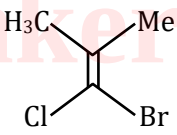
CSI027

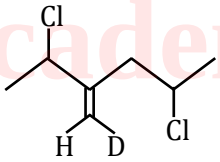
GEOMETRICAL ISOMERISM AND CONFORMATIONS

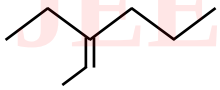
Geometrical Isomerism in Cycloalkanes and Cycloalkenes

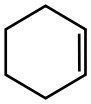
12. Identify molecules which usually can show geometrical isomerism (at room temperature)

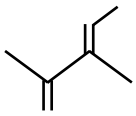
Rankers Academy JEE

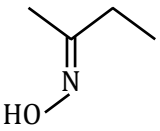
(I) 

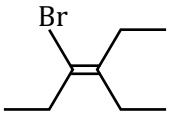
(II) 

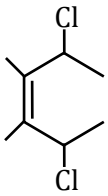
(III) 

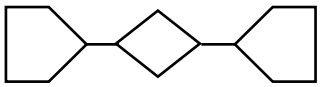
(IV) 

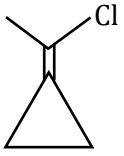
(V) 

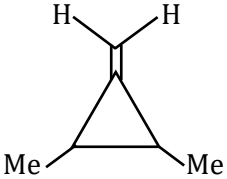
(VI) 

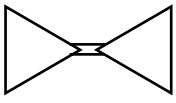
(VII) 

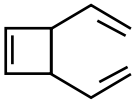
(VIII) 

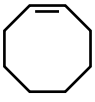
(IX) 

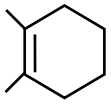
(X) 

(XI) 

(XII) 

(XIII) 

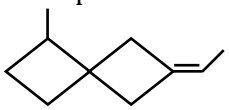
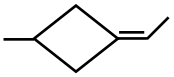
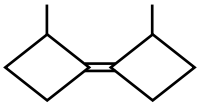
(XIV) 

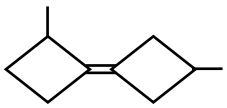
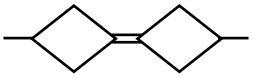
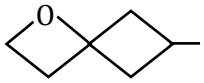
(XV) 

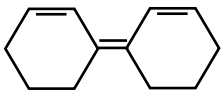
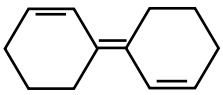
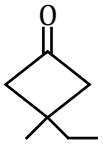
CGI086

For More Join: @IITJEE_Advanced

13. Number of compound which can show geometrical isomerism

(a)  (b)  (c) 

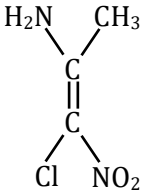
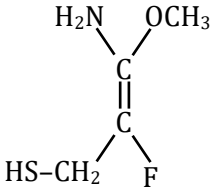
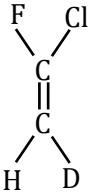
(d)  (e)  (f) 

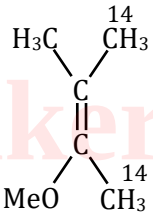
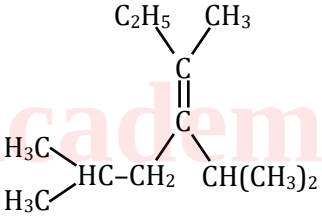
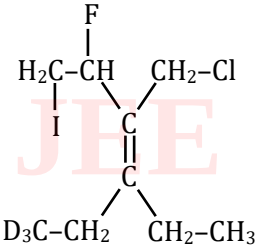
(g)  (h)  (i) 

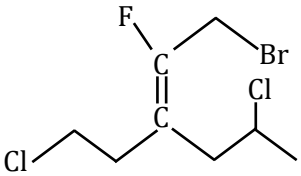
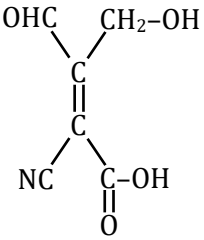
CGI024

Nomenclature of Geometrical Isomers, Cis-Trans Nomenclature, CIP Rules, E-Z Nomenclature and Syn-Anti Nomenclature

14. How many of the following have E configurations:

(i)  (ii)  (iii) 

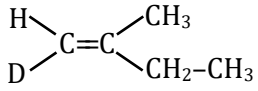
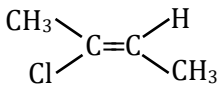
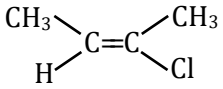
(iv)  (v)  (vi) 

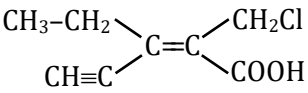
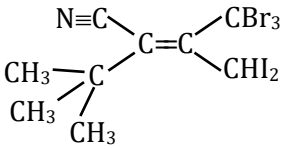
(vii)  (viii) 

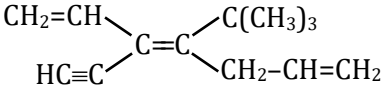
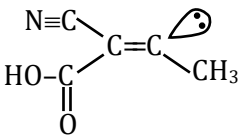
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CGI102

15. Number of compounds having z-configuration:

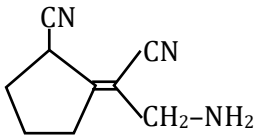
(i)  (ii)  (iii) 

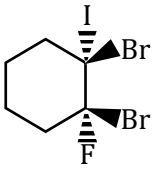
(iv)  (v) 

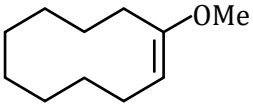
(vi)  (vii) 

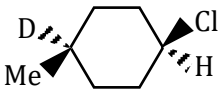
CGI103

For More Join: @IITJEE_Advanced

16. (1) 

(2) 

(3) 

(4) 

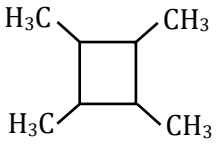
Assign E or Z to the following compounds and write 1 for E and 2 for Z.

Write answer of part (1), (2), (3) & (4) in the same order and present the four digit number as answer in OMR sheet. For example: If all these answer are 9 then fill 9999 in OMR sheet.

CGI022

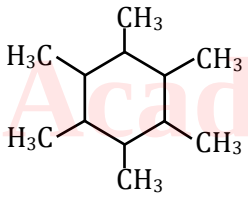
Pseudo Geometrical Isomerism Unit and Calculation of Total Geometrical Isomers

17. How many geometrical isomers are possible for the following compound?



CGI135

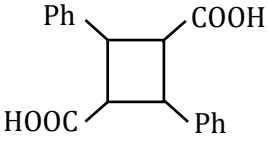
18. How many geometrical isomers are possible for the following compound?



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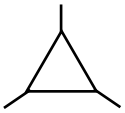
CGI136

19. How many geometrical isomers are possible for the following compound?



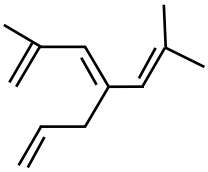
CGI137

20. Number of geometrical isomer for the following compound will be



CGI030

21. Number of geometrical isomers for following compound.

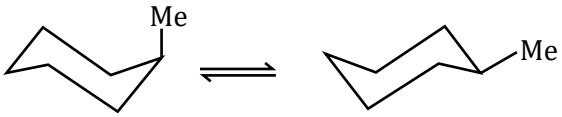


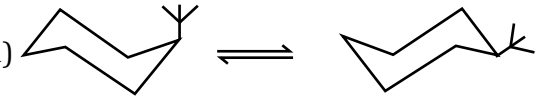
CGI025

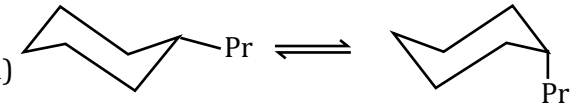
For More Join: @IITJEE_Advanced

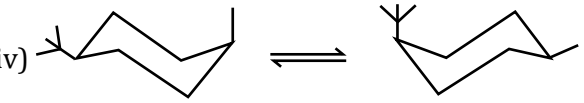
Axial-Equitorial orientation in Cyclo hexane, Ring Flipping; Keq of Flipping; Geometrical Isomerism in Cyclohexane and 1;3 Diaxial Repulsion

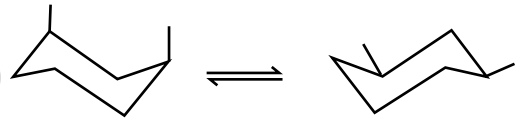
22. The ring flipping equilibrium constant is greater than 1 for

(i) 

(ii) 

(iii) 

(iv) 

(v) 

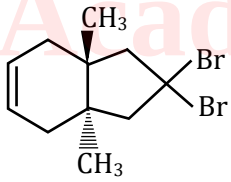
CGI027

OPTICAL ISOMERISM

Chiral Compound and Chiral Centre and asymmetric centre and pseudo chiral centre

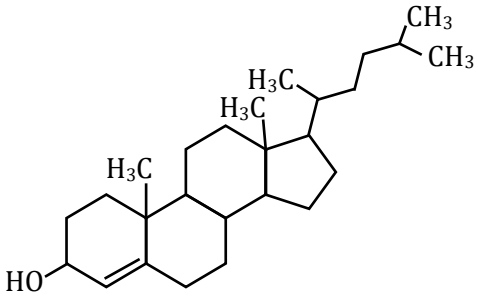
23. How many chiral centres are present in the following compound?

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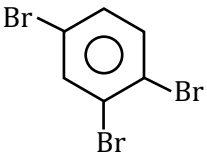
COI141

24. How many chiral centres are present in the following compound?



COI142

Symmetry Elements: Plane of Symmetry

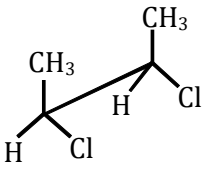
25. 

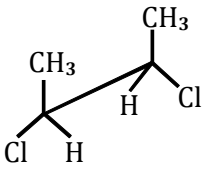
Number of plane of symmetry in above compound is :

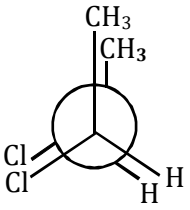
COI098

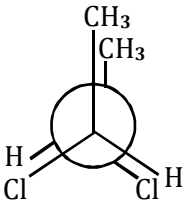
For More Join: @IITJEE_Advanced

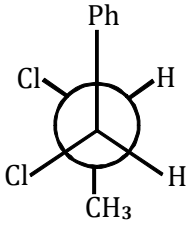
26. Identify the no of compounds containing POS.

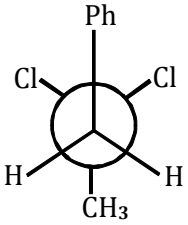
1. 

2. 

3. 

4. 

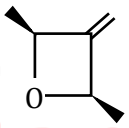
5. 

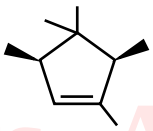
6. 

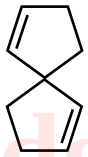
COI099

Optical isomerism in allenes, biphenyls, spiro compounds and in rigid body system

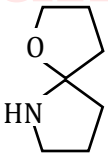
27. How many compounds are optically inactive?

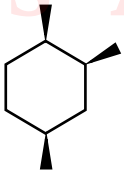
(a) 

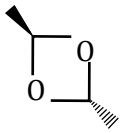
(b) 

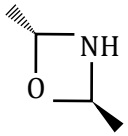
(c) 

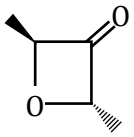
(d) 

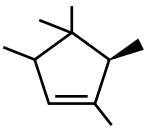
(e) 

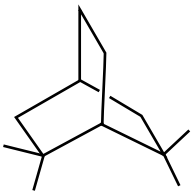
(f) 

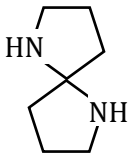
(g) 

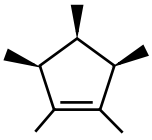
(h) 

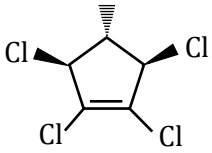
(i) 

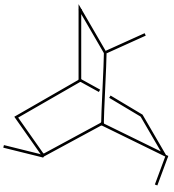
(j) 

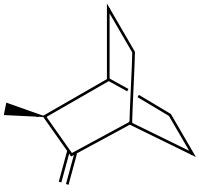
(k) 

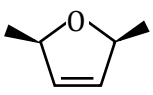
(l) 

(m) 

(n) 

(o) 

(p) 

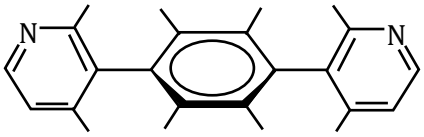
(q) 

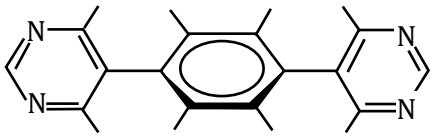
COI132

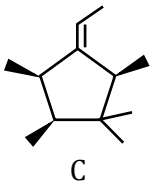
Rankers Academy JEE

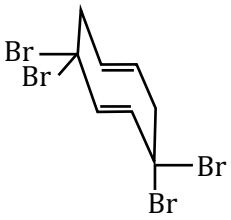
For More Join: @IITJEE_Advanced

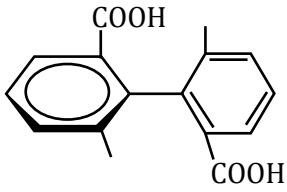
28. How many compounds are optical active?

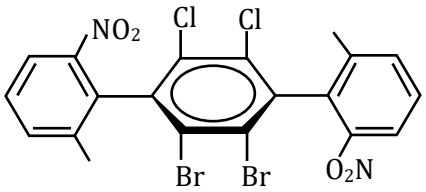
(a) 

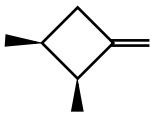
(b) 

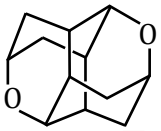
(c) 

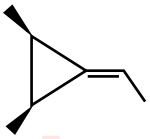
(d) 

(e) 

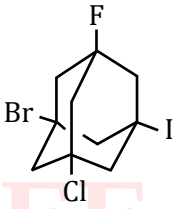
(f) 

(g) 

(h) 

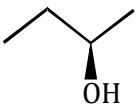
(i) 

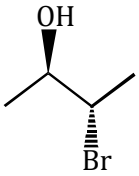
(j) 

(k) 

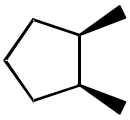
COI133

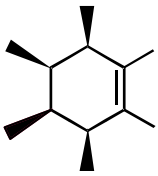
29. How many of the given compounds are chiral:

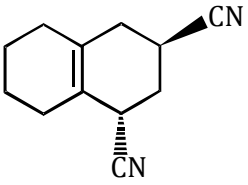
(I) 

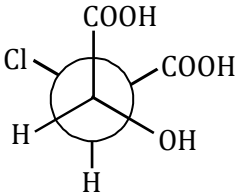
(II) 

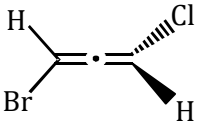
(III) 

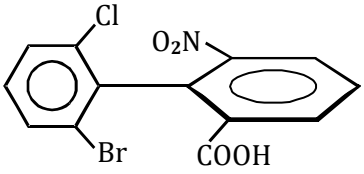
(IV) 

(V) 

(VI) 

(VII) 

(VIII) 

(IX) 

COI026

Optical Purity and Enantiomeric Excess

30. A mixture having 47% d–tartaric acid & 53% ℓ-tartaric acid

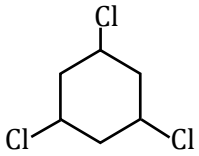
What is the optical purity of the mixture w.r.t. ℓ-isomer :

COI028

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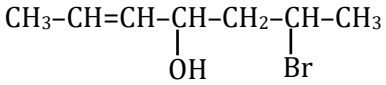
Calculation of Optical Isomers and total isomers

31. Total number of stereoisomers formed by the given compound is:



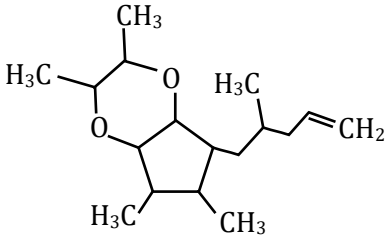
COI130

32. Calculate the total number of stereoisomers for the given compound:



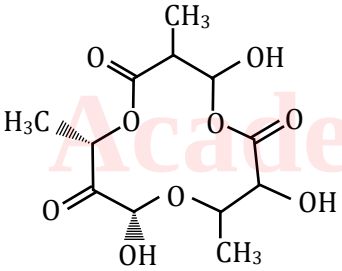
COI131

33. How many stereoisomers are possible for the following compound?



COI134

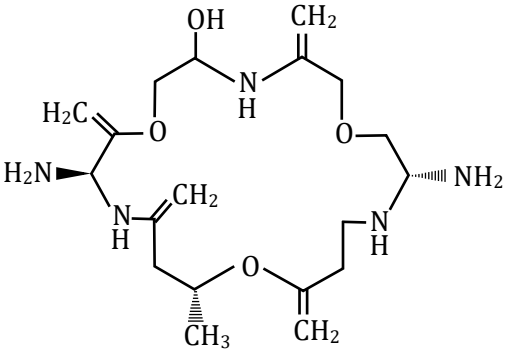
34. How many stereoisomers are possible for the following compound?



Rankers Academy JEE

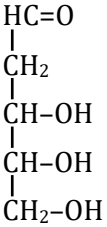
COI135

35. How many stereoisomers are possible for the following compound?



COI137

36. For the given aldopentose how many stereoisomers are possible.



COI029

For More Join: @IITJEE_Advanced

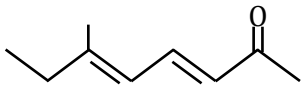
Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

STRUCTURAL ISOMERISM

Enol Content and Kinetic Isotopic Effect

37. Number of H(Hydrogens) will be replaced by D (Deuterium) in given compound when it is kept in mild basic medium for a long time is Z then $\frac{Z}{3}$ is



Deuterium exchanged product.

CSI116

Identification and Number of Structural Isomers

38. Total number of structural isomers of C_9H_{18} containing cyclohexane ring is x then (x–1.5)

CSI114

39. The number of Benzenoid isomers possible for C_6H_3ClBrI is x then $\left(\frac{x}{3}\right)$ is

CSI115

Rankers Academy JEE

GEOMETRICAL ISOMERISM AND CONFORMATIONS

Pseudo Geometrical Isomerism Unit and Calculation of Total Geometrical Isomers

40. Total number of G.I. in following compounds is x then $\frac{x}{2}$.

Write answer as the sum of all the values obtained :

(i) $CH_3–CH=CH–CH_3$

(ii) $CH_3–CH=CH–CH=CH_2$

(iii) $CH_3–CH=CH–CH=NH$

(iv) $CH_3–CH=CH–CH=CH–CH_3$

(v) $CH_3–CH=CH–CH=CH–CH_2–Cl$

(vi) $CH_3–CH=C=CH–CH=CH–CH_2–CH=NH$

CGI149

Questions on Stability and Dipole moment

41. $\mu_{obs} = \Sigma\mu_i x_i$; where μ_i is the dipole moment of stable conformer and x_i is the mole fraction of that conformer of $Z–CH_2–CH_2–Z$ in Newmann's projection. If $\mu_{solution}$ or μ_{net} = 1 D and mole fraction of antiform = 0.82, find μ_{Gauche}.

CGI138

For More Join: @IITJEE_Advanced

OPTICAL ISOMERISM

Calculation of Optical Isomers and total isomers

42. X = total number of isomers of C_7H_{14} (only 5-membered ring).

Y = number of optically active isomers of C_7H_{14} (only 5-membered ring).

Then $\frac{Y}{X}$ = ?

COI148

43. Number of pair of enantiomers possible for 1-bromo- 2-chlorocyclohexane if X.

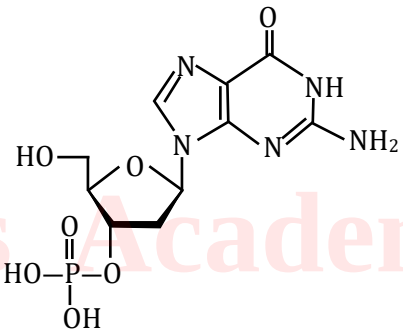
Number of pair of enantiomers possible for 1-chloro- 2-fluorocyclobutane if Y.

Then find out the value of X + Y :

COI030

44. X = stereoisomers possible for the following compound,

Y = Number of chiral centers. Then $\frac{Y}{X}$ = ?



Rankers Academy JEE

COI149

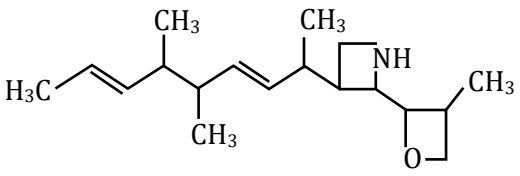
45. X = Total number of isomers for $C_4H_6Br_2$ containing cyclobutane ring (including stereoisomer), Y = number of optically active isomers for $C_4H_6Br_2$ containing cyclobutane ring (including stereoisomer)

Then $\frac{X}{Y}$ = ?

COI150

46. If X = stereoisomers are possible for the following compound,

Then $\frac{X}{100}$ = ?



COI151

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

STRUCTURAL ISOMERISM

1. Which of the following pairs of compounds are positional isomers:- **[JEE (Main) 2015]**

(1) $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$ and $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$

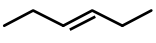
(2) $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$ and $CH_3-\overset{\overset{CH_3}{|}}{CH}-CH_2-CHO$

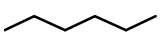
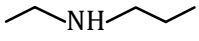
(3) $CH_3-CH_2-CH_2-CH_2-CHO$ and $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CHO$

(4) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$ and $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-CH_2-CHO$

CSI036

2. Given below are two statements :

Statement (I) :  and  are isomeric compounds.

Statement (II) :  NH_2 and  are functional group isomers.

In the light of the above statements, choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) Both **Statement I** and **Statement II** are false
(2) Both **Statement I** and **Statement II** are true
(3) **Statement I** is true but **Statement II** is false
(4) **Statement I** is false but **Statement II** is true

CSI122

3. Compound with molecular formula C_3H_6O can show: **[JEE (Main) 2021]**
(1) Positional isomerism
(2) Both positional isomerism and metamerism
(3) Metamerism
(4) Functional group isomerism

CSI087

4. Monocarboxylic acids are functional isomers of : **[JEE (Main) 2013]**
(1) Esters (2) Amines (3) Ethers (4) Alcohols

CSI035

5. Match **List - I** with **List - II**. **[JEE (Main) 2024]**

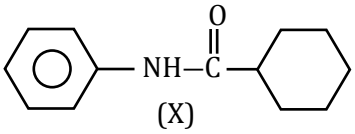
List- I (Pair of Compounds)		**List- II (Isomerism)**	
A.	n-propanol and Isopropanol	I.	Metamerism
B.	Methoxypropane and ethoxyethane	II.	Chain Isomerism
C.	Propanone and propanal	III.	Position Isomerism
D.	Neopentane and Isopentane	IV.	Functional Isomerism

(1) (A)–(II), (B)–(I), (C)–(IV), (D)–(III)
(2) (A)–(III), (B)–(I), (C)–(II), (D)–(IV)
(3) (A)–(I), (B)–(III), (C)–(IV), (D)–(II)
(4) (A)–(III), (B)–(I), (C)–(IV), (D)–(II)

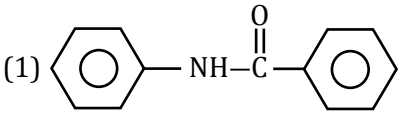
CSI117

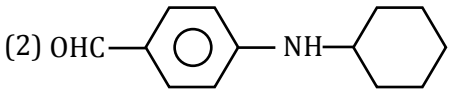
For More Join: @IITJEE_Advanced

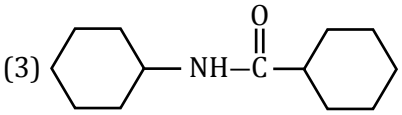
6. Which of the following is metamer of the given compound (X)? **[JEE (Main) 2024]**

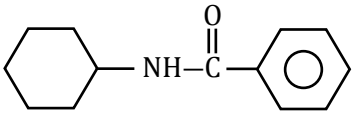


(X)

(1) 

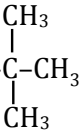
(2) 

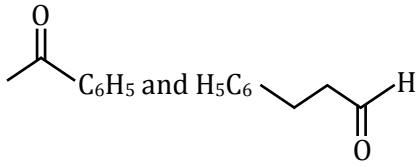
(3) 

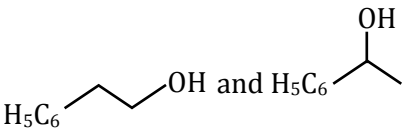
(4) 

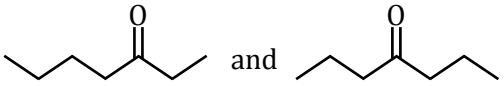
CSI118

7. Which one of the following pairs of isomers is an example of metamerism? **[JEE (Main) 2021]**

(1) $CH_3CH_2CH_2CH_2CH_3$ and 

(2) 

(3) 

(4) 

CSI088

8. Match List-I with List-II. **[JEE (Main) 2023]**

Rankers Academy JEE

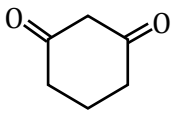
List- I Isomeric pairs		List- II Type of isomers	
A.	Propanamine and N-Methylethanamine	I.	Metamers
B.	Hexan-2-one and Hexan-3-one	II.	Positional isomers
C.	Ethanamide and Hydroxyethanimine	III.	Functional isomers
D.	o-nitrophenol and p-nitrophenol	IV.	Tautomers

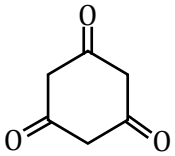
Choose the correct answer from the options given below :-

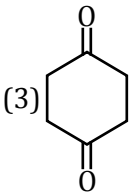
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-I, C-IV, D-II

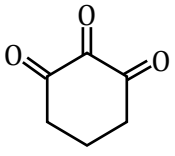
CSI039

9. Highest enol content will be shown by: **[JEE (Main) 2024]**

(1) 

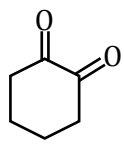
(2) 

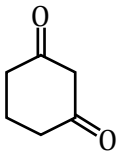
(3) 

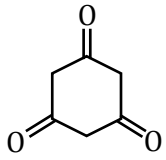
(4) 

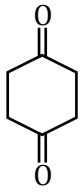
CSI119

10. Which will have the highest enol content? **[JEE (Main) 2022]**

(1) 

(2) 

(3) 

(4) 

CSI089

For More Join: @IITJEE_Advanced

11. Assertion A : Enol form of acetone [CH_3COCH_3] exists in < 0.1% quantity. However, the enol form of acetyl acetone [$CH_3COCH_2OCCH_3$] exists in approximately 15% quantity.
Reason R : enol form of acetyl acetone is stabilized by intramolecular hydrogen bonding, which is not possible in enol form of acetone.
Choose the correct statement : **[JEE (Main) 2021]**
(1) A is false but R is true
(2) Both A and R are true and R is the correct explanation of A
(3) Both A and R are true but R is not the correct explanation of A
(4) A is true but R is false

CSI086

12. The number of different chain isomers for C_7H_{16} is _____. **[JEE (Main) 2024]**

CSI120

13. The total number of monobromo derivatives formed by the alkanes with molecular formula C_5H_{12} is (excluding stereo isomers)___.

[JEE (Main) 2022]

CSI038

14. The number of structural isomers for C_6H_{14} is: **[JEE (Main) 2015]**
(1) 6 (2) 4 (3) 3 (4) 5

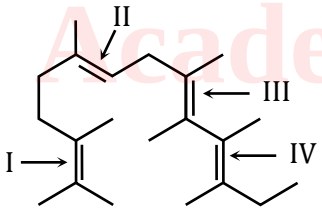
CSI037

15. How many cyclic structures are possible for C_4H_6 :- **[JEE (Main) 2012]**
(1) 3 (2) 5 (3) 4 (4) 6

CSI034

GEOMETRICAL ISOMERISM AND CONFORMATIONS

16. In the following structure, the double bonds are marked as I, II, III and IV



Rankers Academy JEE

Geometrical isomerism is **not** possible at site(s) **[JEE (Main) 2016]**
(1) III (2) I (3) III and IV (4) I and III

CGI037

17. Which of the following compounds will exhibit geometrical isomerism? **[JEE (Main) 2015]**
(1) 2-Phenyl-1-butene (2) 1,1-Diphenyl-1-propene
(3) 1-Phenyl-2-butene (4) 3-Phenyl-1-butene

CGI036

18. Maleic acid and fumaric acids are :- **[AIEEE - 2012]**
(1) Tautomers (2) Chain isomers
(3) Geometrical isomers (4) Functional isomers

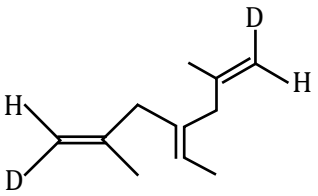
CGI034

19. The **incorrect** statements regarding geometrical isomerism are :
(A) Propene shows geometrical isomerism.
(B) Trans isomer has identical atoms/groups on the opposite sides of the double bond.
(C) Cis-but-2-ene has higher dipole moment than trans-but-2-ene.
(D) 2-methylbut-2-ene shows two geometrical isomers.
(E) Trans-isomer has lower melting point that cis isomer.
Choose the **CORRECT** answer from the options given below : **[JEE (Main) 2025]**
(1) (A), (D) and (E) only (2) (C), (D) and (E) only
(3) (B) and (C) only (4) (A) and (E) only

CGI152

For More Join: @IITJEE_Advanced

20. Number of geometrical isomers possible for the given structure is/are _____.



[JEE (Main) 2024]

CGI150

21. The number of acyclic structural isomers (including geometrical isomers) for pentene are___

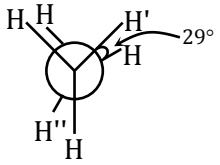
[JEE (Main) 2021]

CGI039

22. The dihedral angle in staggered form of Newman projection of 1, 1, 1-Trichloro ethane is ..... degree. (Round off to the nearest integer) [JEE (Main) 2021]

CGI040

23. In the following skew conformation of ethane, H'–C–C–H" dihedral angle is : [JEE (Main) 2019]



(1) 120° (2) 58° (3) 149° (4) 151°

CGI038

Rankers Academy JEE

24. The incorrect statement regarding conformations of ethane is: [JEE (Main) 2024]

(1) Ethane has infinite number of conformations

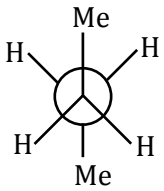
(2) The dihedral angle in staggered conformation is 60°

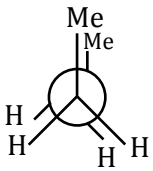
(3) Eclipsed conformation is the most stable conformation.

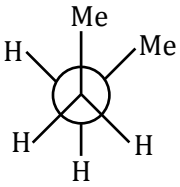
(4) The conformations of ethane are inter-convertible to one-another.

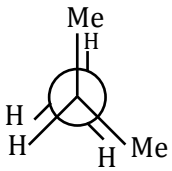
CGI151

25. Which of the following conformations will be the most stable? [JEE (Main) 2023]

(1) 

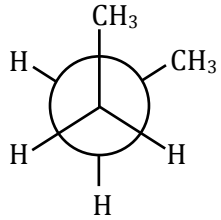
(2) 

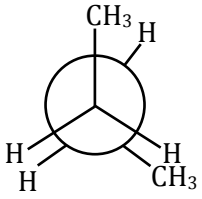
(3) 

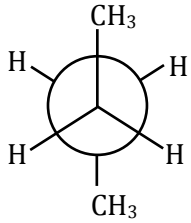
(4) 

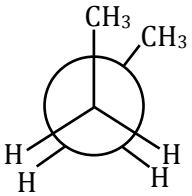
CGI042

26. In the following structures, which one is having staggered conformation with maximum dihedral angle? [JEE (Main) 2022]

(1) 

(2) 

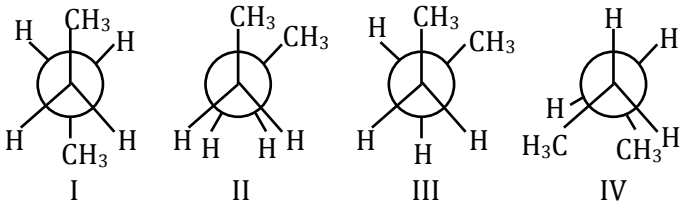
(3) 

(4) 

CGI139

For More Join: @IITJEE_Advanced

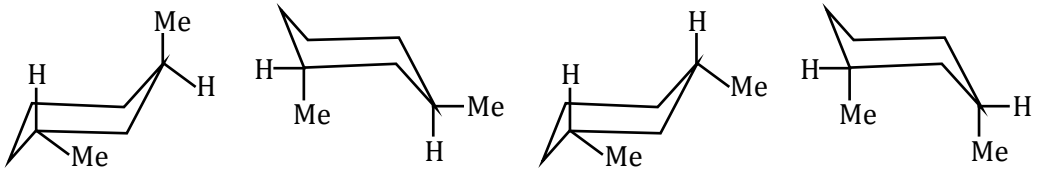
27. Arrange the following conformational isomers of n-butane in order of their increasing potential energy : **[JEE (Main) 2021]**



(1) II < III < IV < I (2) I < IV < III < II (3) II < IV < III < I (4) I < III < IV < II

CGI041

28. Arrange in the correct order of stability (decreasing order) for the following molecules:



[JEE (Main) 2023]

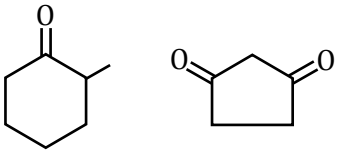
(1) (I) > (II) > (III) > (IV) (2) (IV) > (III) > (II) ≈ (I)
(3) (III) > (I) ≈ (II) > (IV) (4) (I) > (II) ≈ (III) > (IV)

CGI035

OPTICAL ISOMERISM

Pankers Academy JEE

29. Total number of compounds with Chiral carbon atoms from following is ___. **[JEE (Main) 2024]**



$CH_3-CH_2-CH(NO_2)-COOH$

$CH_3-CH_2-CHBr-CH_2-CH_3$

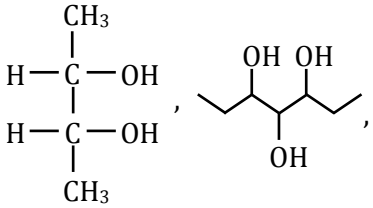
$CH_3-CH(I)-CH_2-NO_2$

$CH_3-CH_2-CH(OH)-CH_2OH$

$CH_3-\underset{\underset{I}{|}}{CH}-CH(I)-C_2H_5$

COI152

30. Total number of optically active compounds from the following is _____. **[JEE (Main) 2024]**



$CH_3-CH_2-CH_2-CH_2-OH$,

$CH_3-CH_2-\underset{\underset{Cl}{|}}{CH}-CH_3$

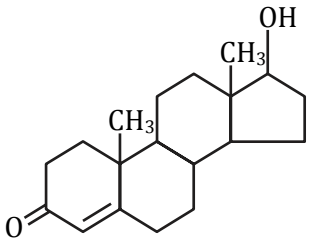
$CH_3-CH_2-CH_2-CH_2-Cl$,

$(CH_3)_2CH-CH_2-CH_2-Cl$

COI153

For More Join: @IITJEE_Advanced

31. Testosterone, which is a steroidal hormone, has the following structure. **[JEE (Main) 2023]**

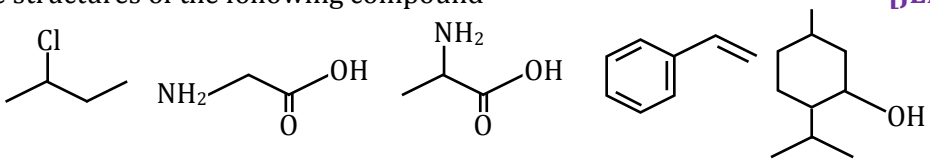


Testosterone

The total number of asymmetric carbon atom/s in testosterone is _____

COI147

32. Observe structures of the following compound **[JEE (Main) 2022]**



The total number of structures/compounds which possess asymmetric carbon atoms is _____.

COI046

33. The optically inactive compound from the following is **[JEE (Main) 2015]**

(1) 2-chloropropanal (2) 2-chlorobutane
(3) 2-chloro-2-methylbutane (4) 2-chloropentane

COI043

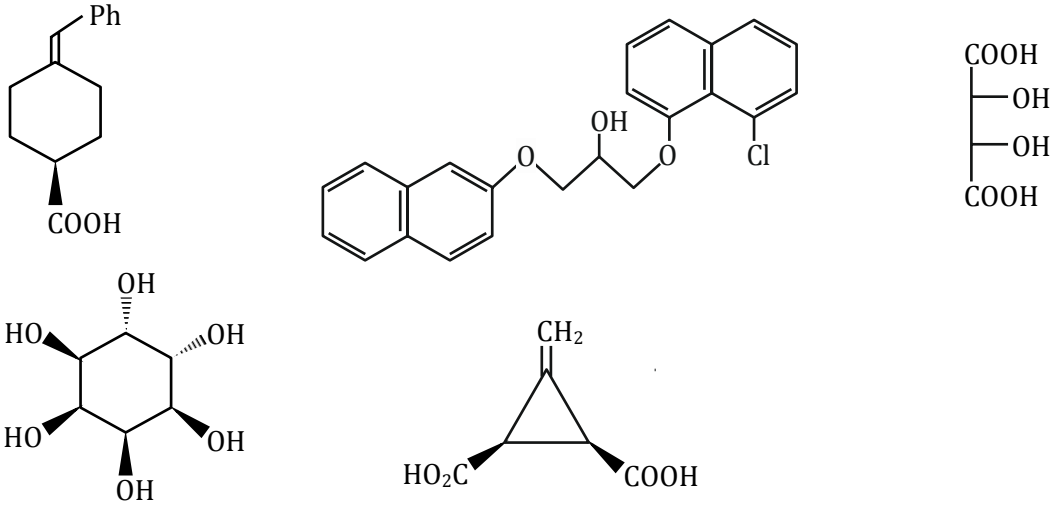
34. Out of the following, the alkene that exhibits optical isomerism is :- **[AIEEE 2010]**

(1) 2-methyl-2-pentene (2) 3-methyl-2-pentene
(3) 4-methyl-1-pentene (4) 3-methyl-1-pentene

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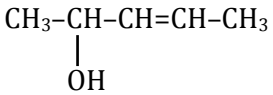
COI041

35. The total number of chiral compound/s from the following is _____. **[JEE (Main) 2023]**



COI146

36. How many different stereoisomers are possible for the given molecule? **[JEE (Main) 2025]**

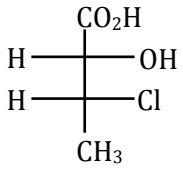


(1) 3 (2) 1 (3) 2 (4) 4

COI161

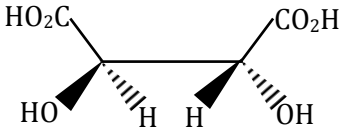
For More Join: @IITJEE_Advanced

37. The absolute configuration of: **[JEE (Main) 2016]**



(1) (2R, 3R) (2) (2R, 3S) (3) (2S, 3R) (4) (2S, 3S)

COI044

38. The absolute configuration of  is **[AIEEE - 2008]**

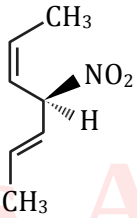
(1) S, S (2) R, R (3) R, S (4) S, R

COI039

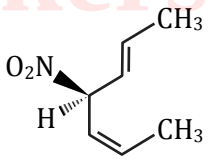
39. Optical activity of an enantiomeric mixture is +12.6° and the specific rotation of (+) isomer is +30°. The optical purity is___% **[JEE (Main) 2022]**

COI143

40. Given below are two statements. **[JEE (Main) 2022]**

Statement I : The compound  is optically active.

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Statement II :  is mirror image of above compound A.

In the light of the above statement, choose the **most appropriate** answer from the options given below.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

COI144

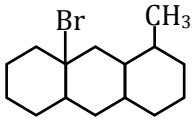
41. The possible number of stereoisomers for 5-phenylpent-4-en-2-ol is _____. **[JEE (Main) 2025]**

COI162

42. Number of optical isomers possible for 2-chlorobutane ..... **[JEE (Main) 2024]**

COI154

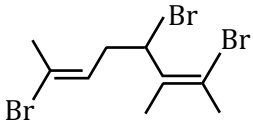
43. The number of optical isomers in following compound is : ____. **[JEE (Main) 2024]**



COI155

For More Join: @IITJEE_Advanced

44. Total number of stereo isomers possible for the given structure: **[JEE (Main) 2024]**



(1) 8 (2) 2 (3) 4 (4) 3

COI156

45. The isomeric deuterated bromide with molecular formula C_4H_8DBr having two chiral carbon atoms is **[JEE (Main) 2023]**

(1) 2–Bromo–1–deuterobutane
(2) 2–Bromo–2–deuterobutane
(3) 2–Bromo–3–deuterobutane
(4) 2–Bromo–1–deutero–2–methylpropane

COI145

46. Total number of isomers (including stereoisomers) obtained on monochlorination of methylcyclohexane is _____. **[JEE (Main) 2022]**

COI047

47. The number of stereoisomers possible for 1,2-dimethyl cyclopropane is : **[JEE (Main) 2021]**

(1) One (2) Four (3) Two (4) Three

COI045

48. The number of stereoisomers possible for a compound of the molecular formula CH_3–CH=CH–(OH)–Me is :- **[AIEEE - 2009]**

(1) 4 (2) 6 (3) 3 (4) 2

COI040

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

STRUCTURAL ISOMERISM

1. The correct combination of names for isomeric alcohols with molecular formula $C_4H_{10}O$ is/are- **[JEE (Advanced) 2014]**

(A) *tert*-butanol and 2-methylpropan-2-ol
(B) *tert*-butanol and 1, 1-dimethylethan-1-ol
(C) *n*-butanol and butan-1-ol
(D) isobutyl alcohol and 2-methylpropan-1-ol

CSI061

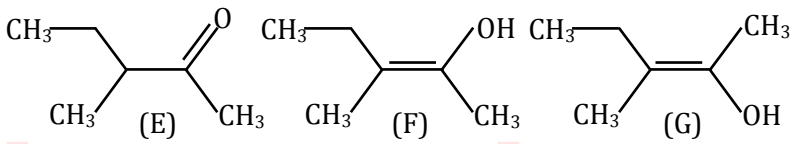
2. The number of structural isomers of C_6H_{14} is : **[IIT - 2007]**

(A) 3 (B) 4 (C) 5 (D) 6

CSI060

GEOMETRICAL ISOMERISM AND CONFORMATIONS

3. The correct statement(s) concerning the structures E, F and G is (are)

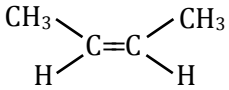


[IIT - 2008]

(A) E, F and G are resonance structures (B) E, F and E, G are tautomers
(C) F and G are geometrical isomers (D) F and G are diastereomers

CGI065

4. Which of the following has the lowest dipole moment : **[IIT - 2002]**

(A) 

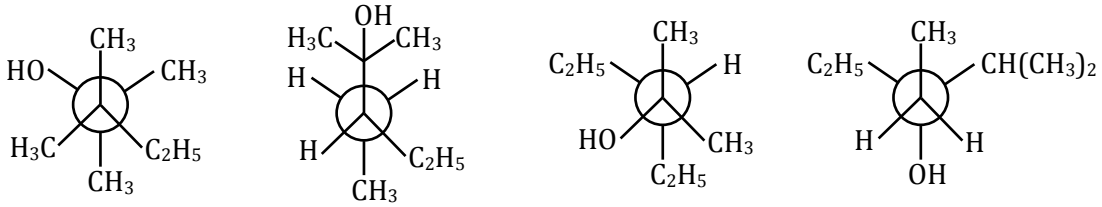
(B) $CH_3–C\equiv C–CH_3$

(C) $CH_3CH_2C\equiv CH$

(D) $CH_2=CH–C\equiv CH$

CGI063

5. Newman projections **P, Q, R** and **S** are shown below : **[JEE (Advanced) 2020]**



Which one of the following options represents identical molecules?

(A) P and Q (B) Q and S (C) Q and R (D) R and S

CGI070

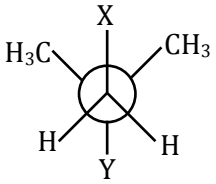
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For More Join: @IITJEE_Advanced

6. The bond energy (in kcal mol^{-1}) of a C–C single bond is approximately: **[IIT - 2010]**

(A) 1 (B) 10 (C) 100 (D) 1000

CGI066

7. In the Newman projection for 2,2–dimethylbutane **[IIT - 2010]**

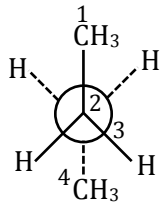


X and Y can respectively be –

(A) H and H (B) H and C_2H_5 (C) C_2H_5 and H (D) CH_3 and CH_3

CGI067

8. If C_2 in below compound is rotated by 120° angle in anticlockwise direction along C_2–C_3, which of the following form will be produced : **[IIT - 2004]**

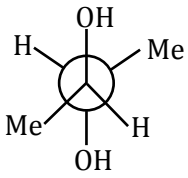


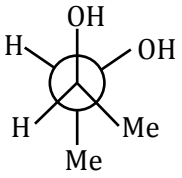
(A) Partial eclipsed (B) Perfectly eclipsed
(C) Perfectly staggered (D) Gauche conformation

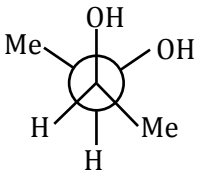
CGI064

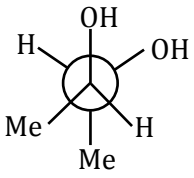
Rankers Academy JEE

9. Among the following, the conformation that corresponds to the most stable conformation of *meso*–butane–2,3–diol is **[JEE (Advanced) 2021]**

(A) 

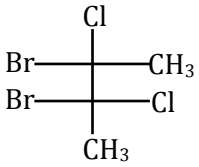
(B) 

(C) 

(D) 

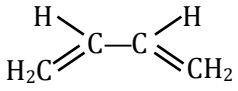
CGI071

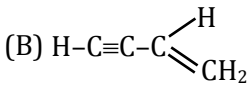
10. The total number(s) of stable conformers with non-zero dipole moment for the following compound is (are) **[JEE (Advanced) 2014]**



CGI069

11. Amongst the given option, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) - **[IIT - 2011]**

(A) 

(B) 

(C) $H_2C=C=O$ (D) $H_2C=C=CH_2$

CGI068

For More Join: @IITJEE_Advanced

OPTICAL ISOMERISM

12. Statement-I : Molecules that are not superimposable on their mirror images are chiral Because Statement-II : All chiral molecules have chiral centres. **[IIT 2007]**

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

COI071

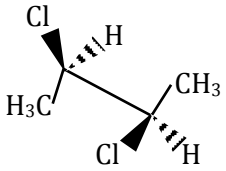
13. The correct statement(s) about the compound given below is (are) :- **[IIT-2008]**

(A) The compound is optically active

(B) The compound possesses centre of symmetry

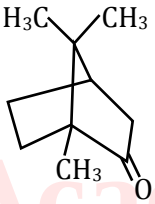
(C) The compound possesses plane of symmetry

(D) The compound possesses axis of symmetry



COI072

14. The total number of stereoisomers that can exist for M is : **[JEE (Advanced) 2015]**

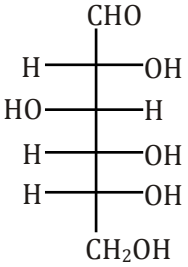


(M)

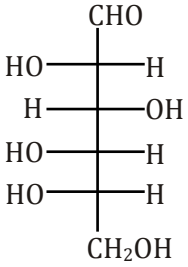
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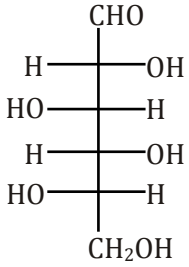
COI078

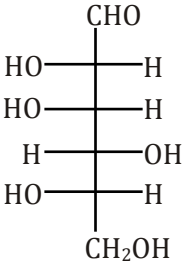
15. The structure of D-(+)-glucose is **[JEE (Advanced) 2015]**

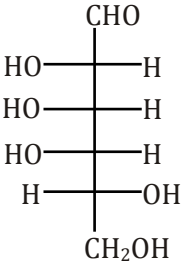


The structure of L(–)-glucose is

(A)


(B)


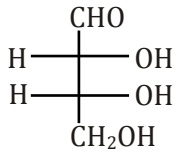
(C)


(D)


COI077

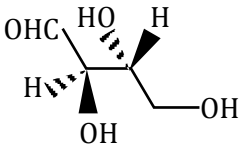
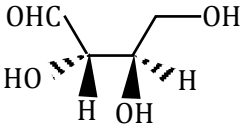
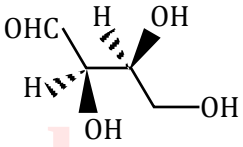
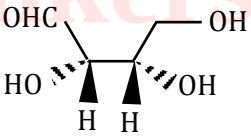
For More Join: @IITJEE_Advanced

16. The Fischer projection of D-erythrose is shown below. **[JEE (Advanced) 2020]**



D–Erythrose

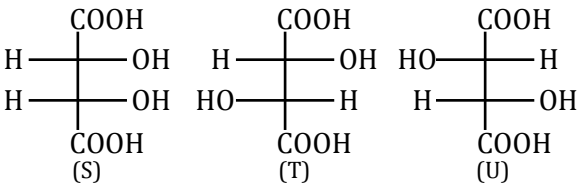
D-Erythrose and its isomers are listed as **P**, **Q**, **R**, and **S** in **Column-I**. Choose the correct relationship of **P**, **Q**, **R**, and **S** with D-erythrose from **Column II**.

	Column–I	**Column–II**
(P)		1. Diastereomer 2. Identical 3. Enantiomer
(Q)		
(R)		
(S)		

(A) P → 2, Q → 3, R → 2, S → 2
(B) P → 3, Q → 1, R → 1, S → 2
(C) P → 2, Q → 1, R → 1, S → 3
(D) P → 2, Q → 3, R → 3, S → 1

COI082

17. P & Q are isomers of dicarboxylic acid $C_4H_4O_4$. Both decolorize Br_2/H_2O, on heating P forms the cyclic anhydride.
Upon treatment with dilute alkaline $KMnO_4$, P as well as Q could produce one or more than one from S, T and U.



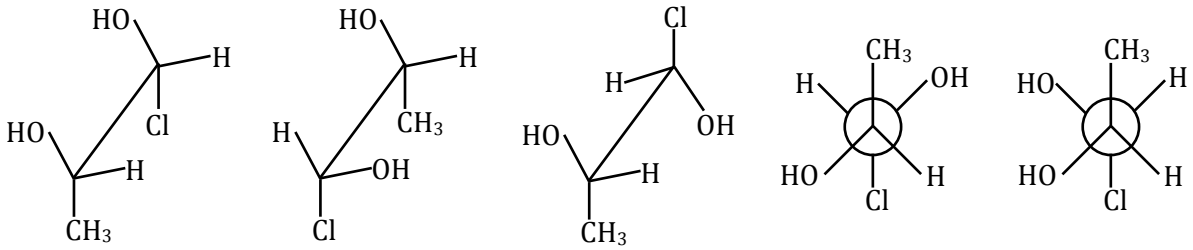
Compounds formed from P and Q are respectively. **[JEE (Advanced) 2013]**
(A) Optically active S and optically active pair (T, U)
(B) Optically inactive S and optically inactive pair (T, U)
(C) Optically active pair (T, U) and optically active S
(D) Optically inactive pair (T, U) and optically inactive S

COI076

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For More Join: @IITJEE_Advanced

18. Which of the given statement(s) about N,O,P and Q with respect to M is (are) correct ? **[IIT 2012]**



(M) (N) (O) (P) (Q)

(A) M and N are non-mirror image stereoisomers

(B) M and O are identical

(C) M and P are enantiomers

(D) M and Q are identical **COI075**

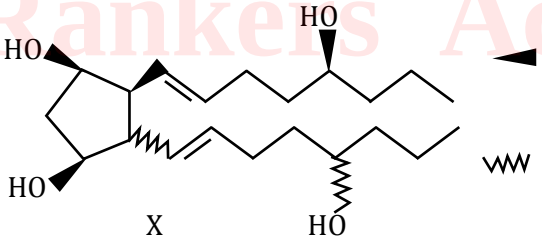
19. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is_ **[JEE (Advanced) 2019]**

COI081

20. For the given compound X, the total number of optically active stereoisomers is___.

[JEE (Advanced) 2018]

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This type of bond indicates that the configuration at the Specific carbon and the geometry of the double bond is fixed.

This type of bond indicates that the configuration at the Specific carbon and the geometry of the double bond is not fixed

COI080

21. The correct statement(s) about the compound $H_3C(HO)HC–CH=CH–CH(OH)CH_3$(X) is are : **[IIT - 2009]**

(A) The total number of stereoisomers possible for X is 6

(B) The total number of diastereomers possible for X is 3

(C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4

(D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2

COI073

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	B	C	B	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	B	B	B	B	D	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	C	B	D	C	A	B	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	C	B	B	B	A	A	A	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	A	C	C	D	B	C	D	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	B	C	C	B	D	D	C	B	D

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AD	BC	ACD	ABD	BC	ABC	ABC	ABC	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AB	BCD	ACD	ABC	BD	ABC	ABC	ABC	AC	ACD
Que.	21	22	23	24	25	26	27			
Ans.	ABCD	CD	D	C	C	C	B			

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EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	5	3	4	5	8	4	5	7	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	6	9	6	5	3	2112	4	8	5	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	2	8	1	3	7	8	6	6
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	2	8	8	64	32	4	3.33	10.5	3.33	11.50
Que.	41	42	43	44	45	46				
Ans.	5.55	0.5	4.00	0.375	3.00	10.24				

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	4	1	4	4	4	4	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	9	8	4	2	2	3	3	1	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	6	60	3	3	1	3	4	3	5	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	6	3	3	4	2	4	3	2	42	3
Que.	41	42	43	44	45	46	47	48		
Ans.	4	2	32	1	3	12	4	1		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ACD	C	BCD	B	C	C	BD	D	B	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	BC	C	AD	2	A	C	B	ABC	10	7
Que.	21									
Ans.	AD									

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